

#

Numerical:

Test data

Threshold = 0.5

Actual label	Predicted Score	Predicted label	0.7	0.4	0.2
1	0.85	1	1 TP	1 TP	1 TP
0	0.2	0	0 TN	0 TN	0 TN
0	0.6	1	0 TN	1 FP	1 FP
1	0.7	1	0 FN	1 TP	1 TP
1	0.4	0	0 FN	1 TP	1 TP
0	0.55	1	0 TN	1 FP	1 FP
1	0.5	0	0 FN	1 TP	1 TP
0	0.65	0	0 TN	1 FP	1 FP
0	0.35	0	0 TN	0 TN	1 FP
0	0.6	1	0 FN	1 TP	1 TP

1. Confusion Matrix

2. Accuracy

3. Precision

4. Recall / Sensitivity

5. F1-Score

6. ROC-Curve (threshold: 0.7, 0.5, 0.4, 0.2)

original +ve original -ve

predicted +ve	3	3
predicted -ve	2	2

threshold = 0.4

TP	FP
5	3
FN	TN
0	2

$$\text{Sensitivity} = 5/5 = 1$$

$$\text{Specificity} = 2/2+3 = 0.4$$

$$\text{Accuracy} = 3+2/10 = 0.5$$

$$\text{Precision} = 3/6 = 0.5$$

$$\text{Recall/Sensitivity} = 3/3+2 = 0.6$$

$$\text{F1 score} = 2 \times \frac{0.5 \times 0.6}{0.5+0.6} = \frac{0.3 \times 2}{1.1} = 0.54$$

$$\text{Specificity} = 2/(2+3) = 0.4$$

0.7
threshold

TP	FP
1	0
FN	TN
4	5

threshold = 0.2

TP	FP
5	4
FN	TN
0	1

$$\text{Sensitivity} = 5/5 = 1$$

$$\text{Specificity} = 1/5 = 0.2$$

$$\text{Sensitivity} = 1/1+0 = 1$$

$$\text{Specificity} = 5/5+0 = 1$$

